Non-Markovian **Quantum Refrigeration in the** Quantum Switch

Quantum Thermodynamics Conference 2026 — Natal, Brazil

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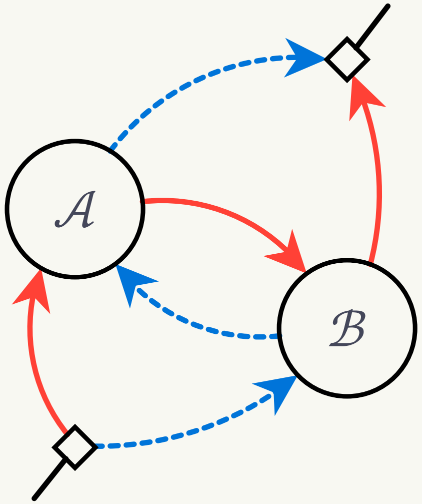
13 August 2026

Quantum Switch

- Composition of quantum channels.
- A *superposition of causal orders.*



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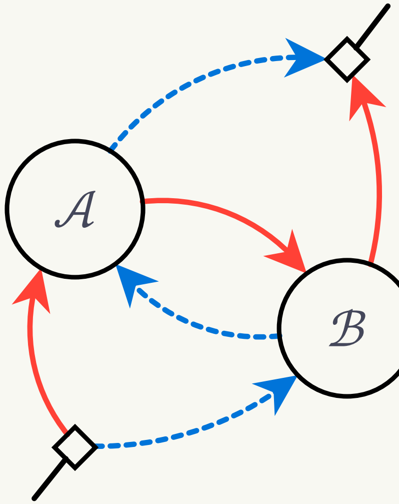
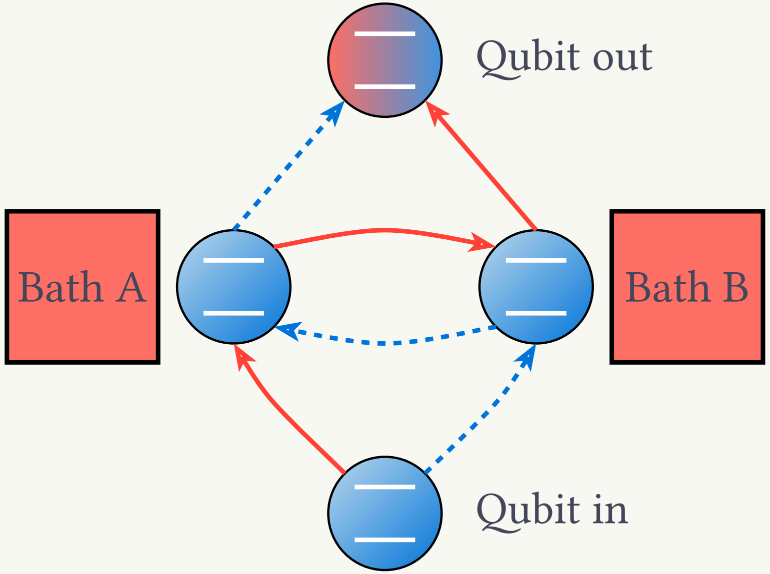




- Different effective temperature than baths.
- Enables thermodynamic tasks not possible with fixed causal orders.

Quantum Switch

Such compositions can grant many different enhancements:

- **Refrigeration**

  1. Felce D, Vedral V (2020) Quantum Refrigeration with Indefinite Causal Order. Phys Rev Lett 125:70603. https://doi.org/10.1103/PhysRevLett.125.070603

- **Work extraction**

  2. Simonov K, Francica G, Guarnieri G, Paternostro M (2022) Work extraction from coherently activated maps via quantum switch. Physical Review A 105:. https://doi.org/10.1103/physreva.105.032217

- **Communication capacity**

  3. Ebler D, Salek S, Chiribella G (2018) Enhanced Communication with the Assistance of Indefinite Causal Order. Physical Review Letters 120:. https://doi.org/10.1103/physrevlett.120.120502

- **Metrology**

  4. Zhao X, Yang Y, Chiribella G (2020) Quantum Metrology with Indefinite Causal Order. Physical Review Letters 124:. https://doi.org/10.1103/physrevlett.124.190503

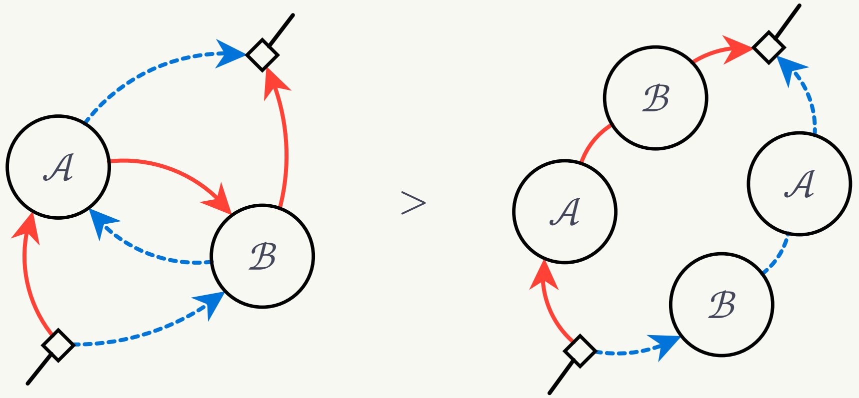
- **Computational complexity**

  5. Araújo M, Costa F, Brukner Č (2014) Computational Advantage from Quantum-Controlled Ordering of Gates. Physical Review Letters 113:. https://doi.org/10.1103/physrevlett.113.250402

Quantum Switch

Origin of these enhancements?

1. **Indefinite Causal Order**



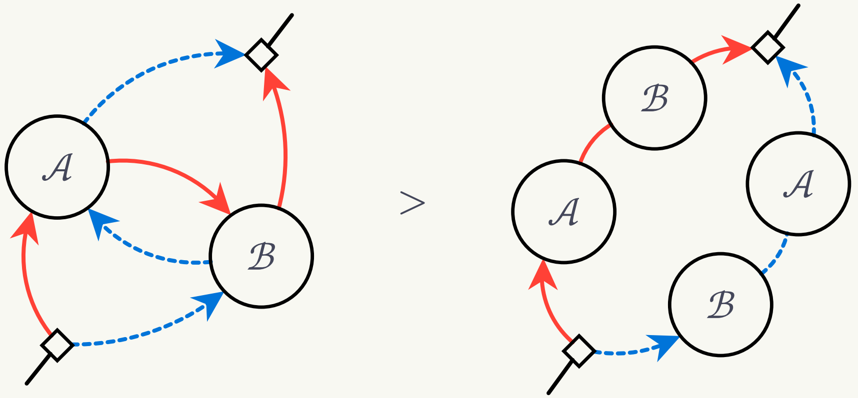
6. Chiribella G, Kristjánsson H (2019) Quantum Shannon theory with superpositions of trajectories. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 475:20180903. https://doi.org/10.1098/rspa.2018.0903

7. Kristjánsson H, Chiribella G, Salek S, et al (2020) Resource theories of communication. New Journal of Physics 22:73014. https://doi.org/10.1088/1367-2630/ab8ef7

Quantum Switch

Origin of these enhancements?

1. **Indefinite Causal Order**



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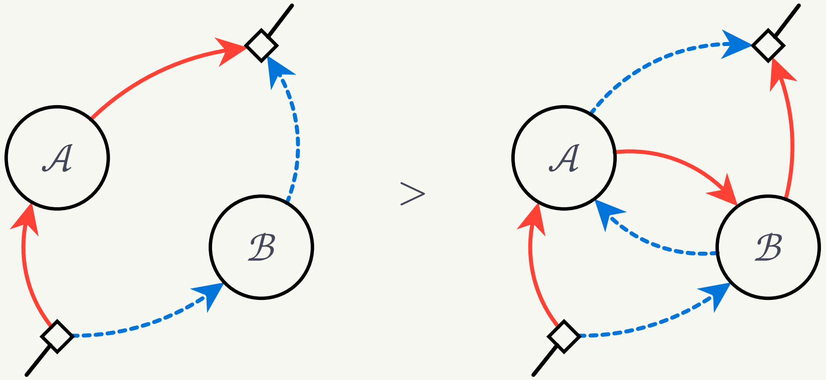
7. Kristjánsson H, Chiribella G, Salek S, et al (2020) Resource theories of communication. New Journal of Physics 22:73014. https://doi.org/10.1088/1367-2630/ab8ef7

2. **Coherent Superposition**



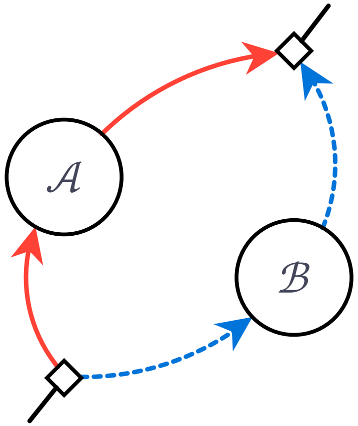
8. Guérin PA, Rubino G, Brukner Ča (2019) Communication through quantum-controlled noise. Physical Review A 99:. https://doi.org/10.1103/physreva.99.062317

9. Abbott AA, Wechs J, Horsman D, et al (2020) Communication through coherent control of quantum channels. Quantum 4:333. https://doi.org/10.22331/q-2020-09-24-333

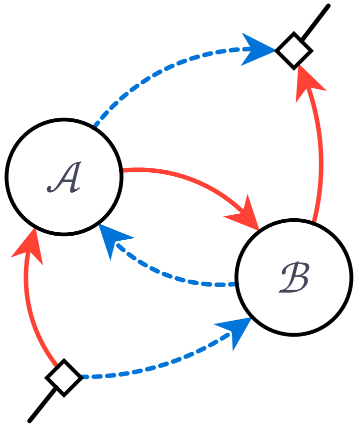
Key Idea

Origin of these enhancements?

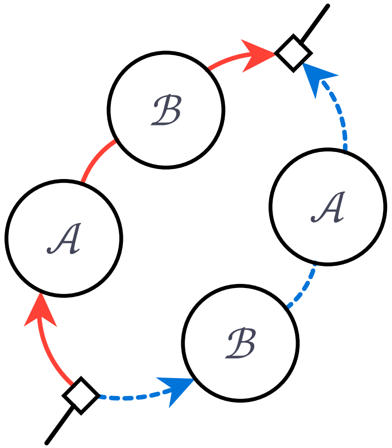
3. **Non-Markovianity**
   - Generalizes all 3 compositions.
   - Offers a dual explanation for quantum switch's enhancements.



Φ^{indep}
1st timestep



Φ^{order}
Fully non-Markovian



Φ^{path}
Fully Markovian

The Quantum Switch

Composition of quantum channels

$$\rho^S \;\text{—}\boxed{\mathcal{A}}\text{—}\; \mathcal{A}(\rho^S) \qquad\qquad \rho^S \;\text{—}\boxed{\mathcal{B}}\text{—}\; \mathcal{B}(\rho^S)$$

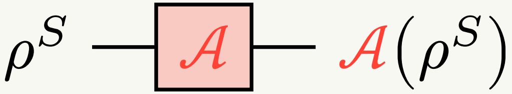
$$\mathcal{A}(\rho^S) = \sum_i A_i \rho^S A_i^\dagger \qquad\qquad \mathcal{B}(\rho^S) = \sum_i B_i \rho^S B_i^\dagger$$

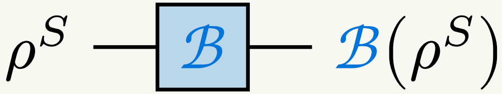
10. *DiVincenzo DP, Shor PW, Smolin JA (1998) Quantum-channel capacity of very noisy channels. Physical Review A 57:830–839. https://doi.org/10.1103/physreva.57.830*

11. *Hastings MB (2009) Superadditivity of communication capacity using entangled inputs. Nature Physics 5:255–257. https://doi.org/10.1038/nphys1224*

12. *Smith G, Yard J (2008) Quantum Communication with Zero-Capacity Channels. Science 321:1812–1815. https://doi.org/10.1126/science.1162242*

Composition of quantum channels

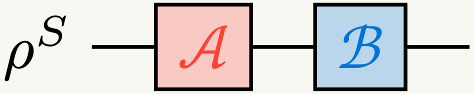




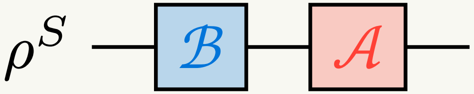
$$\mathcal{A}(\rho^S) = \sum_i A_i \rho^S A_i^\dagger$$

$$\mathcal{B}(\rho^S) = \sum_i B_i \rho^S B_i^\dagger$$

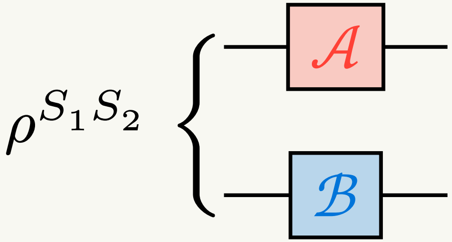
Series composition (*bottleneck inequality*):



or



Parallel composition (*superadditivity/ superactivation*):



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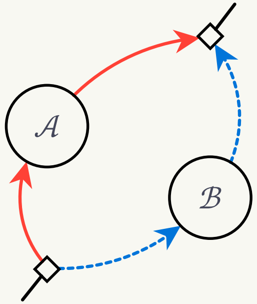
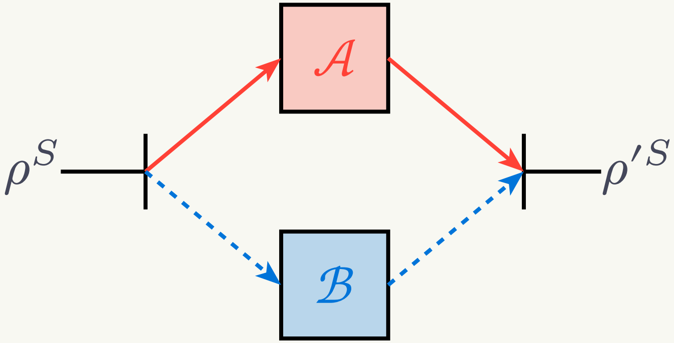
Composition of quantum channels

Superposition of independent channels Φ^{indep}:





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Composition of quantum channels

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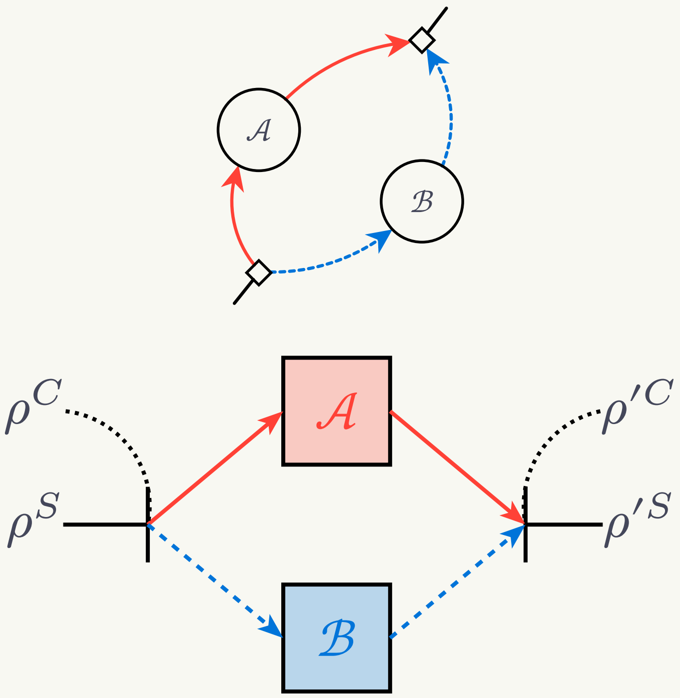

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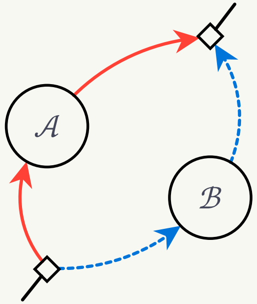

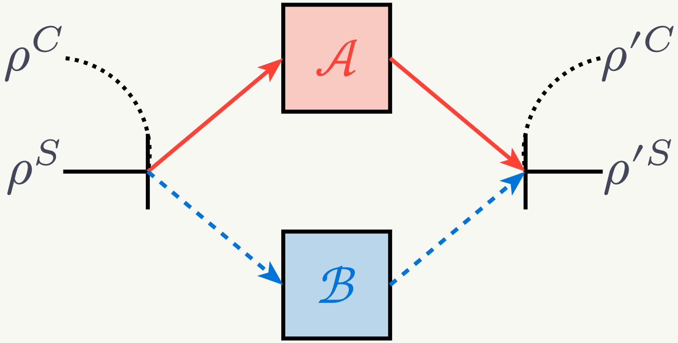

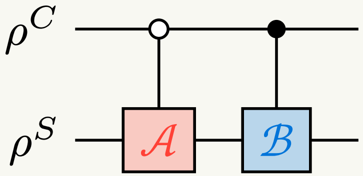

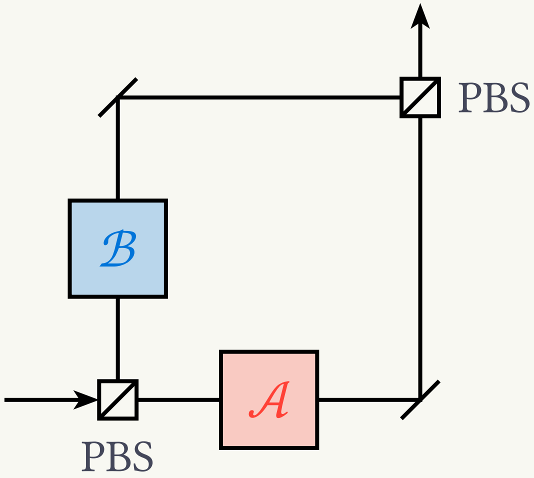

$$\Phi^{\text{indep}}\left(\rho^C \otimes \rho^S\right) = \sum_{i,j} K_{ij}\left(\rho^C \otimes \rho^S\right)K_{ij}^\dagger$$

$$K_{ij} = \eta_i |0\rangle\langle 0|^C \otimes A_i + \eta_j |1\rangle\langle 1|^C \otimes B_j$$

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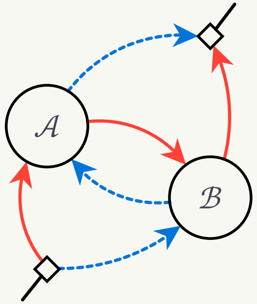
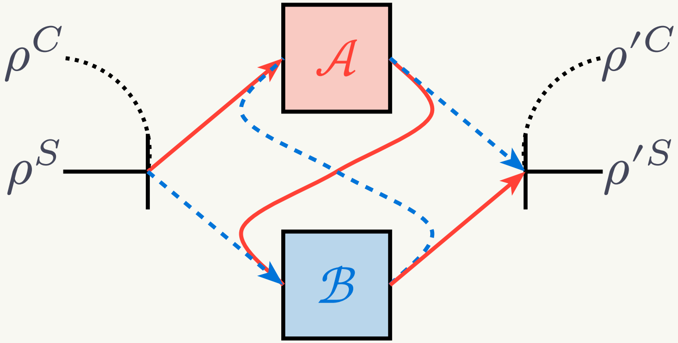
Quantum switch

Superposition of causal orders Φ^{order}:

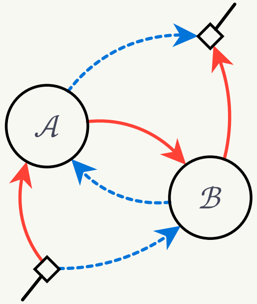


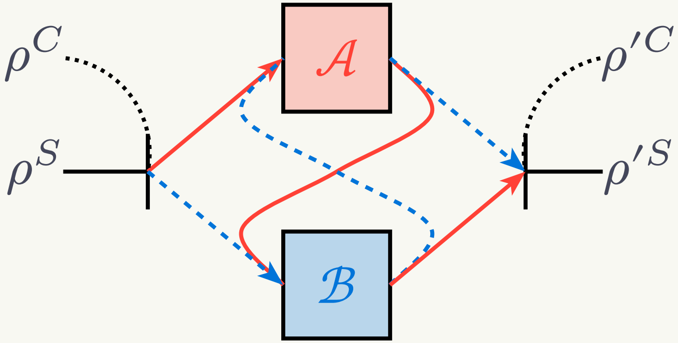


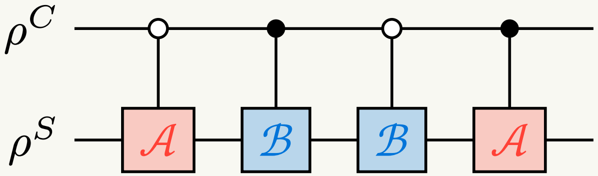
13. Chiribella G, D'Ariano GM, Perinotti P, Valiron B (2013) Quantum computations without definite causal structure. Physical Review A 88:. https://doi.org/10.1103/physreva.88.022318

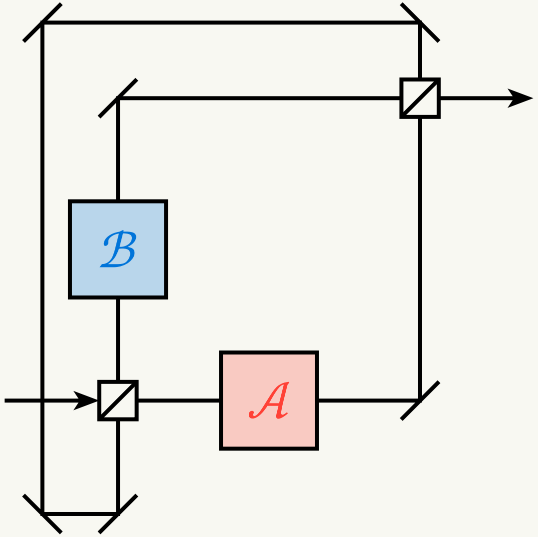
Quantum switch

Superposition of causal orders Φ^{order}:









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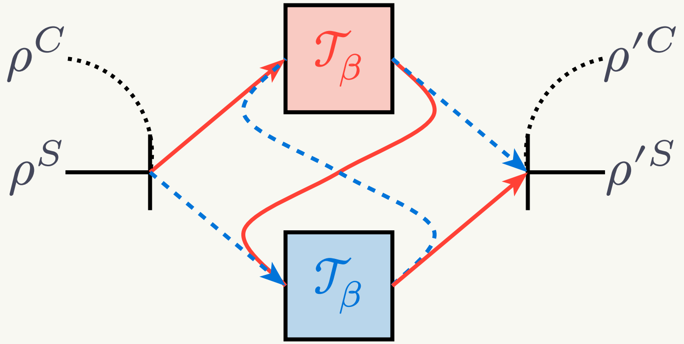
Quantum switch refrigeration example



Fully thermalizing channels:

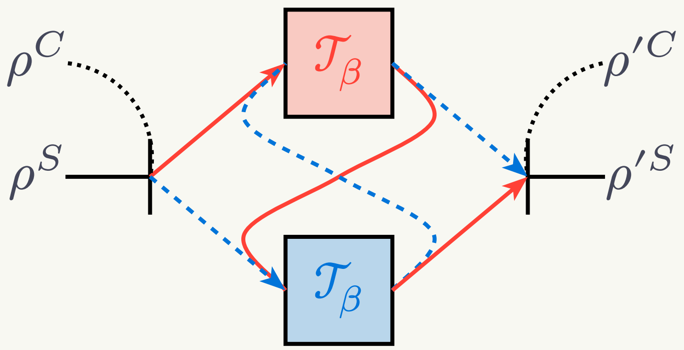
$$\mathcal{T}_\beta(\rho) = \sum_i K_i \rho K_i^\dagger = \sigma_\beta$$

$$K_i = \frac{1}{\sqrt{2}} \sqrt{\sigma_\beta} U_i$$

$$\{U_i\} = \{I, X, Y, Z\}$$

1. *Felce D, Vedral V (2020) Quantum Refrigeration with Indefinite Causal Order. Phys Rev Lett 125:70603. https://doi.org/10.1103/PhysRevLett.125.070603*

Quantum switch refrigeration example



Quantum switch operation:

$$\Phi^{\text{order}}\left(\rho^C \otimes \rho^S\right) = \frac{1}{2}\left[|0\rangle\langle 0|^C + |1\rangle\langle 1|^C\right] \otimes \sigma_\beta$$
$$+\frac{1}{2}\left[|0\rangle\langle 1|^C + |1\rangle\langle 0|^C\right] \otimes \sigma_\beta \rho^S \sigma_\beta$$

Fully thermalizing channels:

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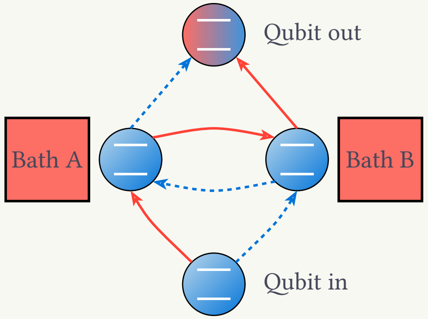
$$K_i = \frac{1}{\sqrt{2}}\sqrt{\sigma_\beta} U_i$$

$$\{U_i\} = \{I, X, Y, Z\}$$

Measure in the $|\pm\rangle^C$ basis:

$$\mathrm{Tr}_C\left[\left(|\pm\rangle\langle\pm|^C \otimes I^S\right)\Phi^{\text{sw}}\left(\rho^C \otimes \rho^S\right)\right]$$

$$= \frac{\sigma_\beta}{2} \pm \frac{\sigma_\beta \rho^S \sigma_\beta}{2}$$



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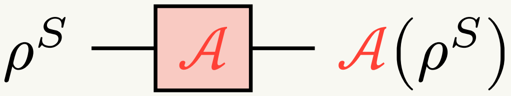
Non-Markovianity in the Quantum Switch

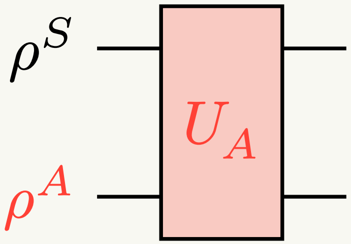
Environmental representation

Open system:



$$\mathcal{A}(\rho^S) = \sum_i A_i \rho^S A_i^\dagger$$

Unitary:



$$\mathcal{A}(\rho^S) = \mathrm{Tr}_A\left[U_A(\rho^S \otimes \rho^A)U_A^\dagger\right]$$

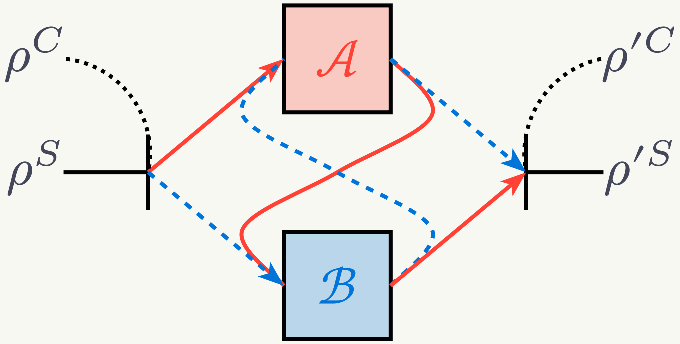
$\rho^S \otimes \rho^A \Rightarrow$ CPTP (sufficient but not necessary condition)

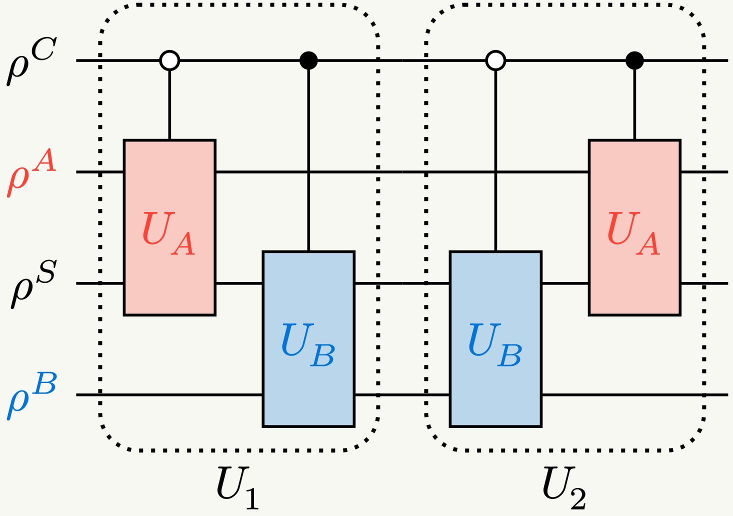
14. *Buscemi F (2014) Complete Positivity, Markovianity, and the Quantum Data-Processing Inequality, in the Presence of Initial System-Environment Correlations. Physical Review Letters 113:. https://doi.org/10.1103/physrevlett.113.140502*

15. *Brodutch A, Datta A, Modi K, et al (2013) Vanishing quantum discord is not necessary for completely positive maps. Physical Review A 87:. https://doi.org/10.1103/physreva.87.042301*

16. *Dominy JM, Shabani A, Lidar DA (2015) A general framework for complete positivity. Quantum Information Processing 15:465–494. https://doi.org/10.1007/s11128-015-1148-0*

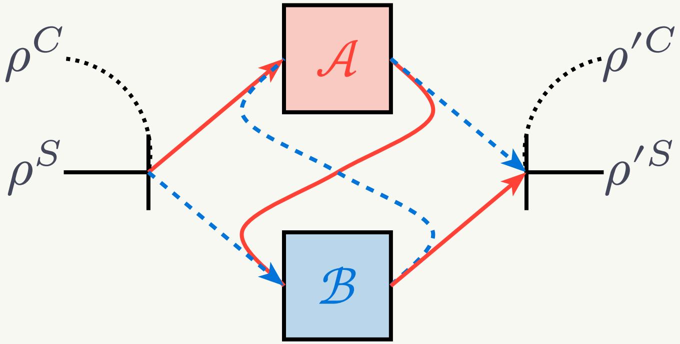
Environmental representation

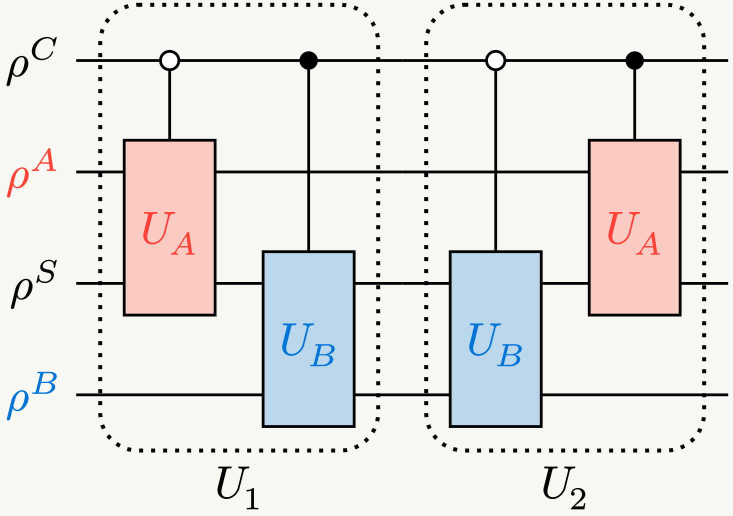




17. Cheong JW, Pradana A, Chew LY (2022) Communication advantage of quantum compositions of channels from non-Markovianity. Phys Rev A 106:52410. https://doi.org/10.1103/PhysRevA.106.052410

Environmental representation





Quantum switch operation:

$$\Phi^{\text{order}}\left(\rho^C \otimes \rho^S\right)$$

$$= \mathrm{Tr}_{A,B}\left[U_2 U_1 \left(\rho^C \otimes \rho^A \otimes \rho^S \otimes \rho^B\right) U_1^\dagger U_2^\dagger\right]$$

$$U_1 = |0\rangle\langle 0|^C \otimes U_A \otimes I^B + |1\rangle\langle 1|^C \otimes I^A \otimes U_B$$

$$U_2 = |0\rangle\langle 0|^C \otimes I^A \otimes U_B + |1\rangle\langle 1|^C \otimes U_A \otimes I^B$$

17. *Cheong JW, Pradana A, Chew LY (2022) Communication advantage of quantum compositions of channels from non-Markovianity. Phys Rev A 106:52410. https://doi.org/10.1103/PhysRevA.106.052410*

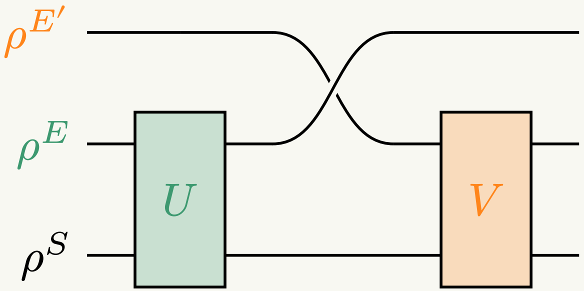
Non-Markovianity

Markovian if CP-divisible:

$$\Phi_{t_0\to t_2} = \Phi_{t_1\to t_2} \circ \Phi_{t_0\to t_1}, \quad \forall t_0 < t_1 < t_2$$

$$\Rightarrow \Phi_{t_0\to t_1} = \mathrm{Tr}_E\left[U\left(\rho^S \otimes \rho^E\right)U^\dagger\right]$$

$$\Rightarrow \Phi_{t_1\to t_2} = \mathrm{Tr}_{E'}\left[V\left(\rho^S \otimes \rho^{E'}\right)V^\dagger\right]$$

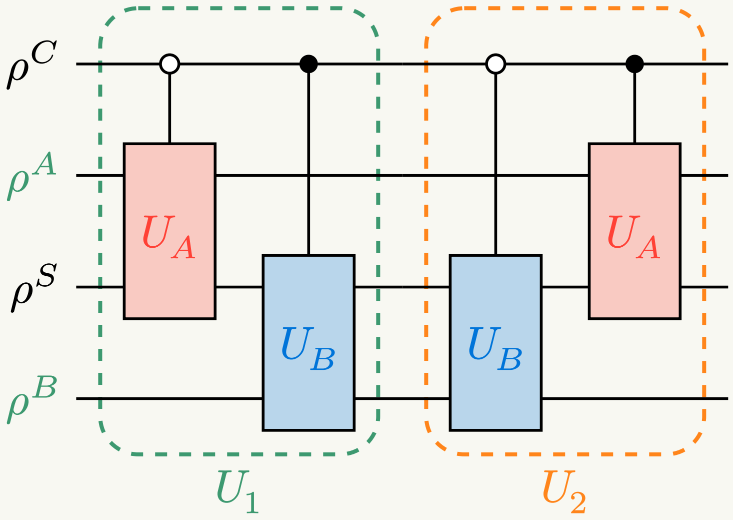


> Markovianity requires **2 uncorrelated environments**.

18. *Lindblad G (1976) On the generators of quantum dynamical semigroups. Communications in Mathematical Physics 48:119–130. https://doi.org/10.1007/bf01608499*

19. *Gorini V (1976) Completely positive dynamical semigroups of N-level systems. Journal of Mathematical Physics 17:821. https://doi.org/10.1063/1.522979*

Non-Markovianity



$$\Phi^{\text{order}} = \Gamma^{\text{order}}_{t_1 \to t_2} \circ \Phi^{\text{order}}_{t_0 \to t_1}$$

$$\Phi^{\text{order}}_{t_0 \to t_1} = \mathrm{Tr}_{A,B}\left[U_1\left(\rho^C \otimes \rho^A \otimes \rho^S \otimes \rho^B\right)U_1^\dagger\right]$$

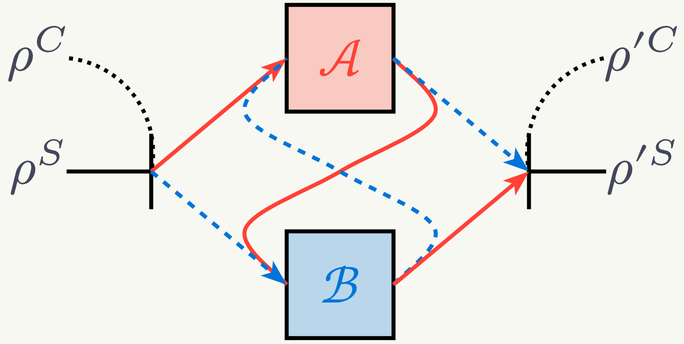
$$\Gamma^{\text{order}}_{t_1 \to t_2} = \mathrm{Tr}_{A,B}\left[U_2\left(\rho^{CASB}\right)U_2^\dagger\right]$$

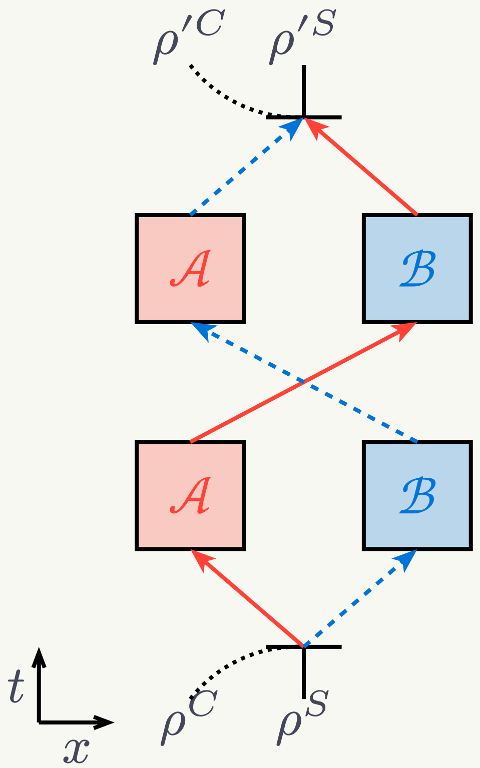
Not CP-divisible in general, can have **intrinsic non-Markovianity**.

- Also can have non-Markovian **backflow of information**.

17. *Cheong JW, Pradana A, Chew LY (2022) Communication advantage of quantum compositions of channels from non-Markovianity. Phys Rev A 106:52410. https://doi.org/10.1103/PhysRevA.106.052410*

Non-Markovianity





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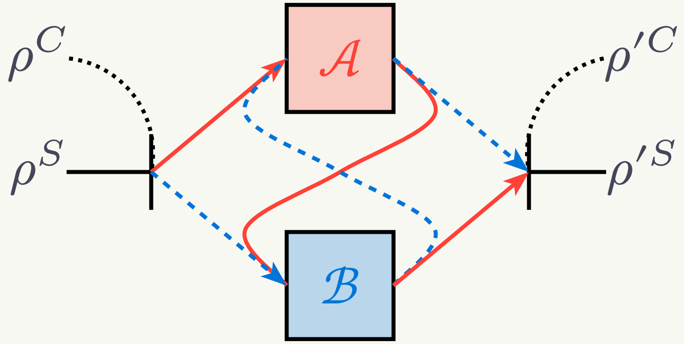
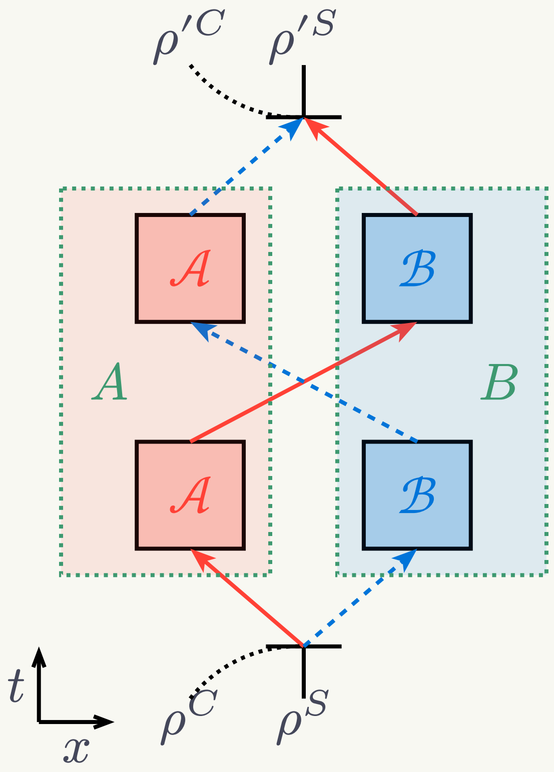
Non-Markovianity

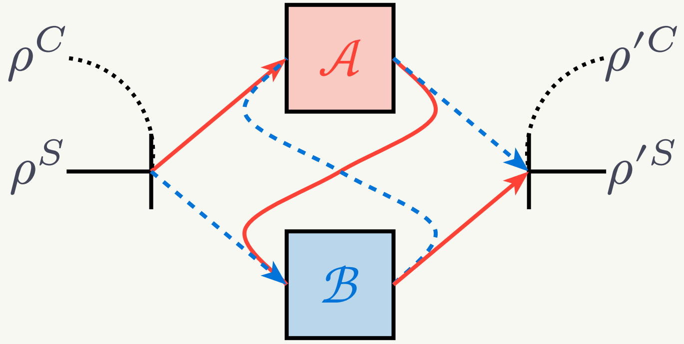


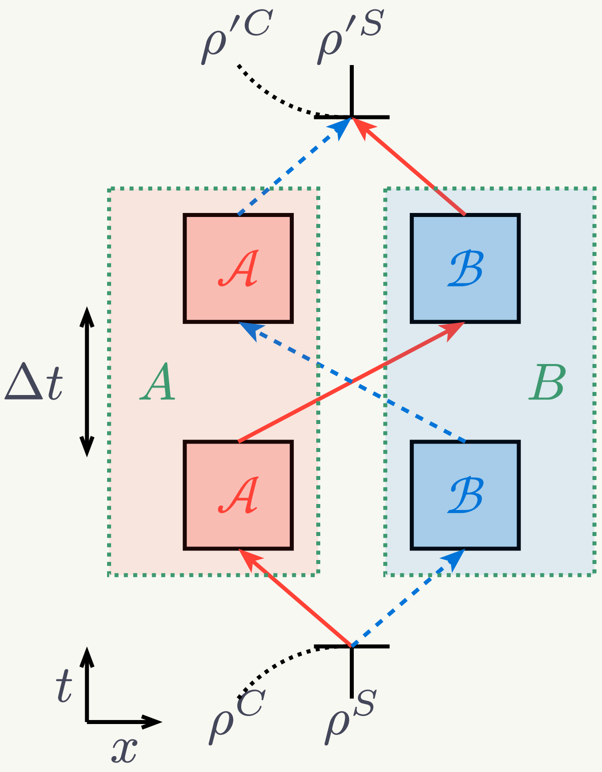


Coherence-activated non-Markovianity.
Not simply a superposition of non-Markovian channels.

17. *Cheong JW, Pradana A, Chew LY (2022) Communication advantage of quantum compositions of channels from non-Markovianity. Phys Rev A 106:52410. https://doi.org/10.1103/PhysRevA.106.052410*

Non-Markovianity





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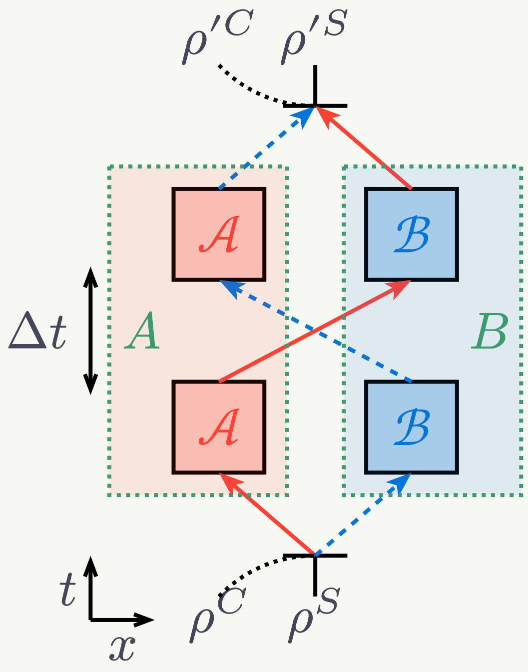
Controlling non-Markovian memory



$\gamma \neq 1$

Non-Markovian

Generalized amplitude damping channel:

$$K_1 = \sqrt{1-N}\left(|0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|\right)$$

$$K_2 = \sqrt{\gamma(1-N)}|0\rangle\langle 1|$$

$$K_3 = \sqrt{N}\left(\sqrt{1-\gamma}|0\rangle\langle 0| + |1\rangle\langle 1|\right)$$

$$K_4 = \sqrt{\gamma N}|1\rangle\langle 0|$$

$$N = \langle 1|\sigma_\beta|1\rangle,$$

$$0 \leq \gamma \leq 1$$

$$W_{ij} = I^C \otimes K_i^A \otimes I^S \otimes K_j^B$$

$$\rho^{CA'SB'} = \sum_{ij} W_{ij}\rho^{CASB}W_{ij}^\dagger$$

20. *Cheong JW, Pradana A, Chew LY (2023) Effects of non-Markovianity on daemonic ergotropy in the quantum switch. Physical Review A 108:. https://doi.org/10.1103/physreva.108.012201*

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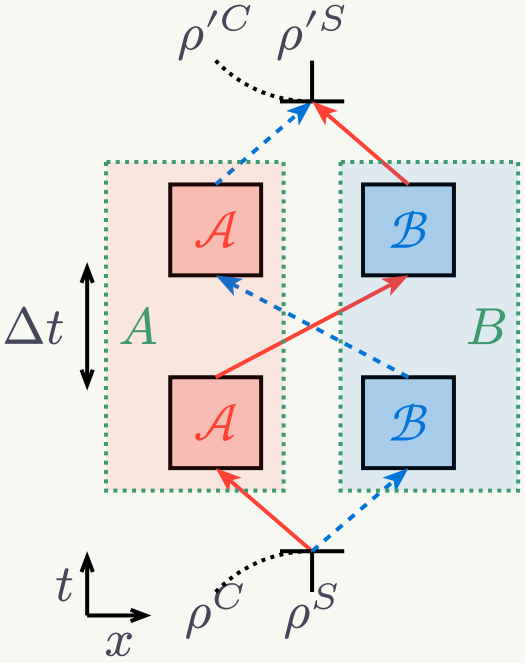

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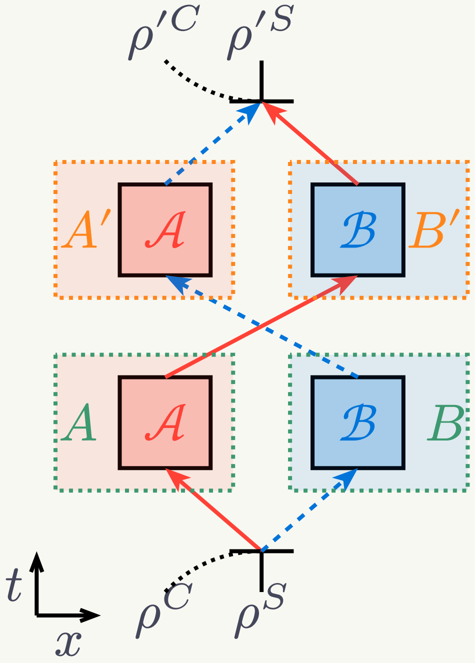

$\gamma = 1$

Markovian

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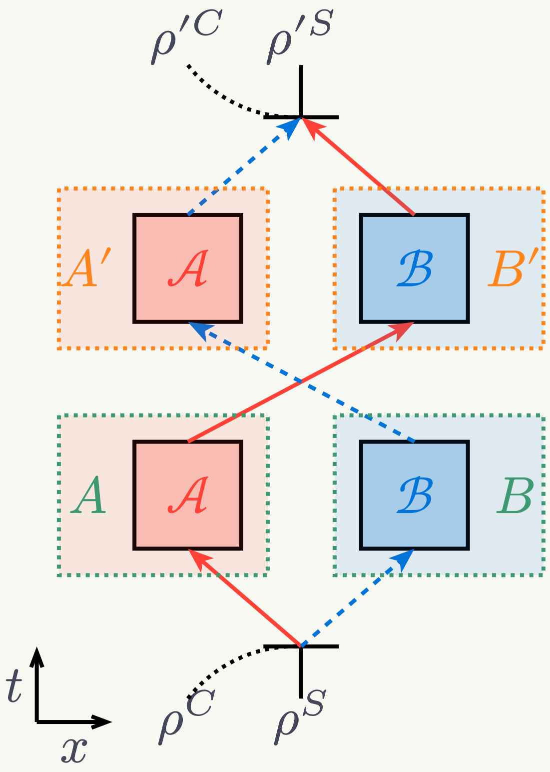
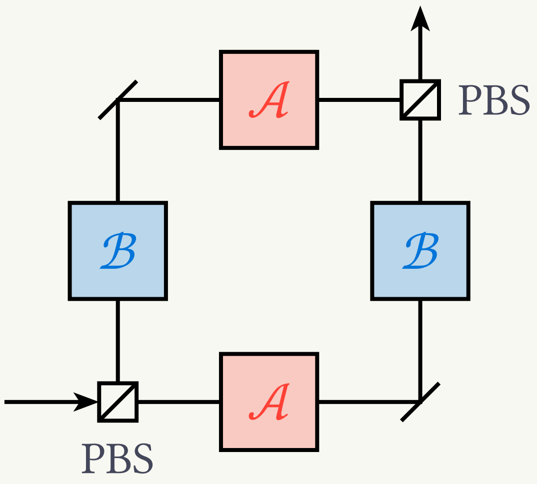
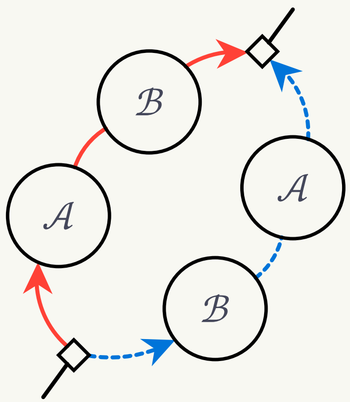
Controlling non-Markovian memory

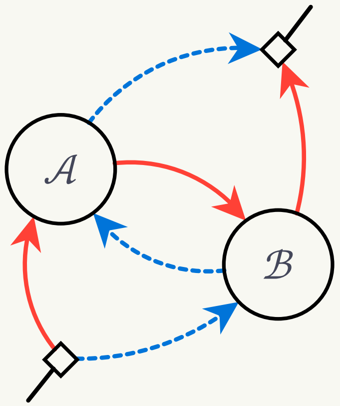






Fully Markovian case reduces to
superposition of paths Φ^{path}.

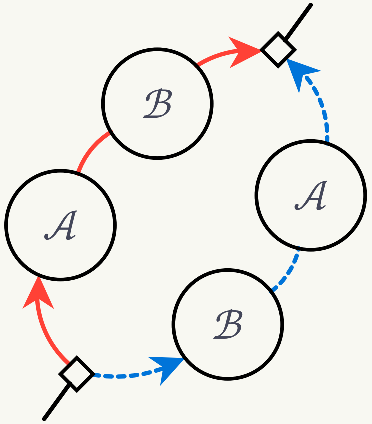
Non-Markovianity is the answer?



Φ^{order}

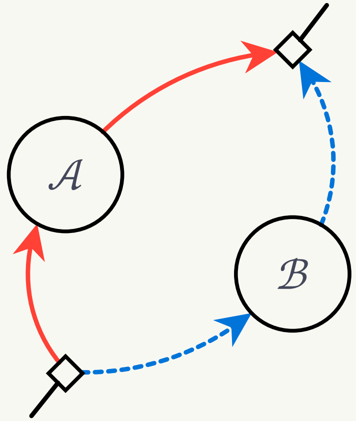
Non-Markovian

$>$



Φ^{path}

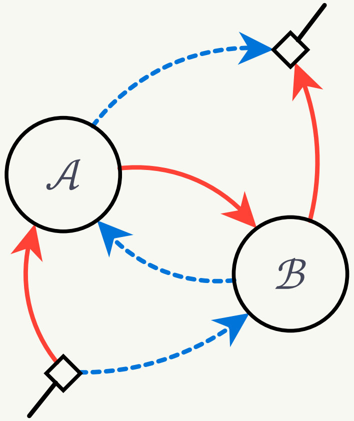
Markovian



Φ^{indep}

First timestep of Φ^{order} and Φ^{path}

$>$



Φ^{order}

Non-Markovian

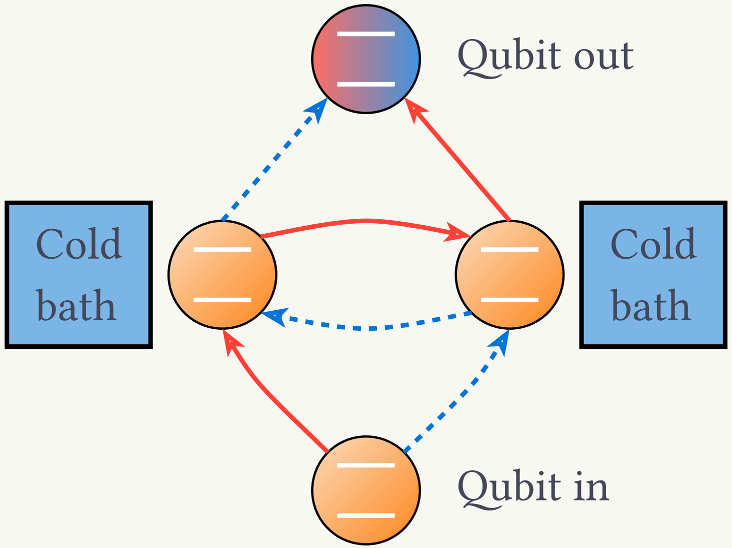
Non-Markovianity grants an advantage.

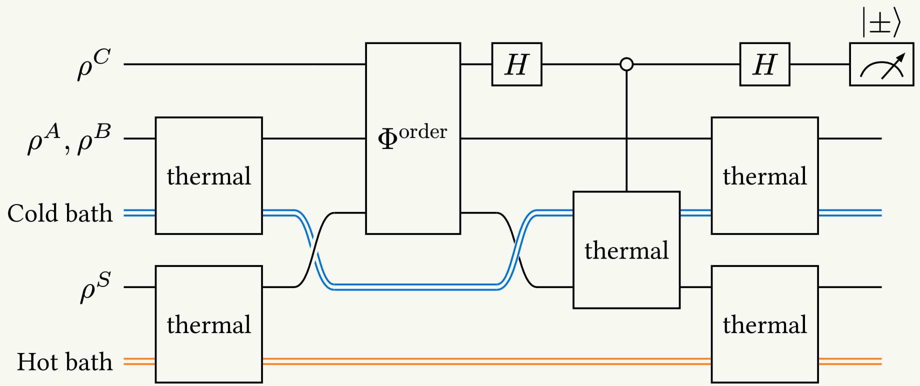
- Typically from backflow of information.

Non-Markovianity does not **grant an advantage.**

- Typically without backflow of information.

Non-Markovian quantum refrigeration

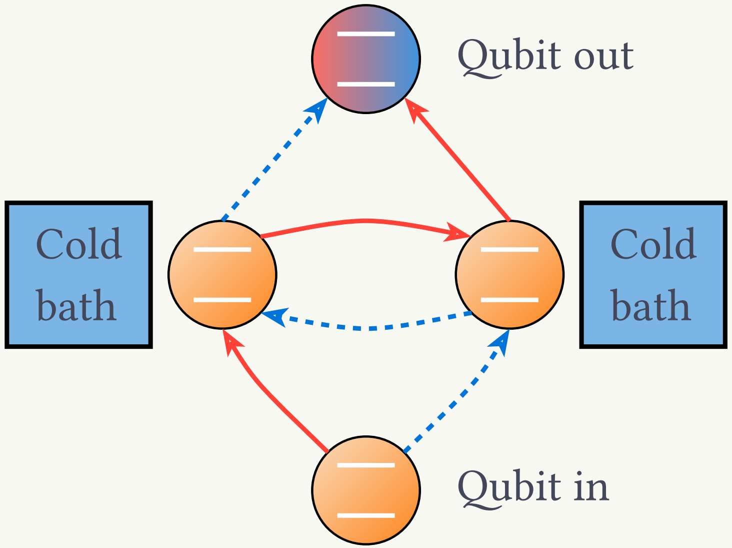


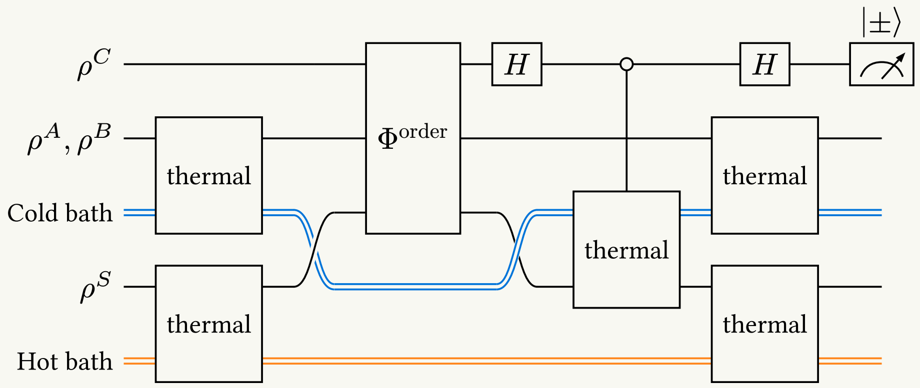


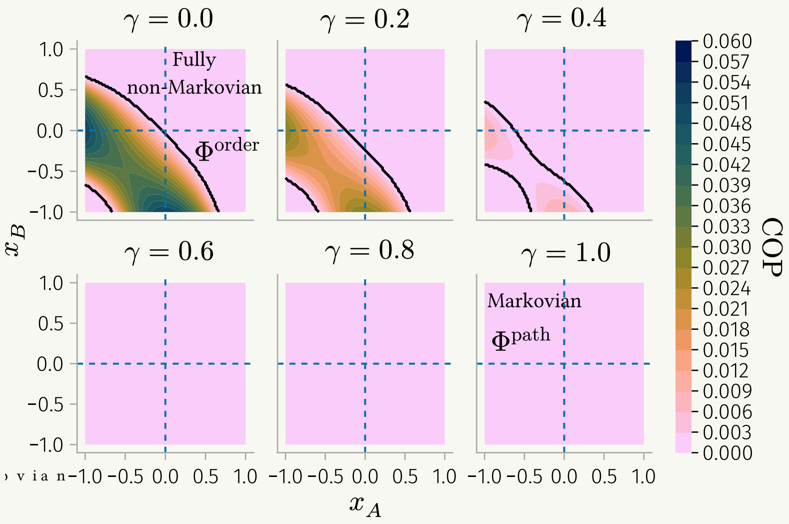
- Refrigeration cycle with an initially hotter qubit,
- compute Coefficient of Performance (COP),
- vary amount of non-Markovian memory.

21. *Cheong JW, Pradana A, Chew LY (2024) Non-Markovian refrigeration and heat flow in the quantum switch. Phys Rev A 110:22220. https://doi.org/10.1103/PhysRevA.110.022220*

Non-Markovian quantum refrigeration







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Effects of non-Markovianity

Non-Markovian enhancements in the quantum switch Φ^{order} shown for:

- **Refrigeration**

  21. Cheong JW, Pradana A, Chew LY (2024) Non-Markovian refrigeration and heat flow in the quantum switch. Phys Rev A 110:22220. https://doi.org/10.1103/PhysRevA.110.022220

- **Work extraction**

  20. Cheong JW, Pradana A, Chew LY (2023) Effects of non-Markovianity on daemonic ergotropy in the quantum switch. Physical Review A 108:. https://doi.org/10.1103/physreva.108.012201

- **Communication capacity**

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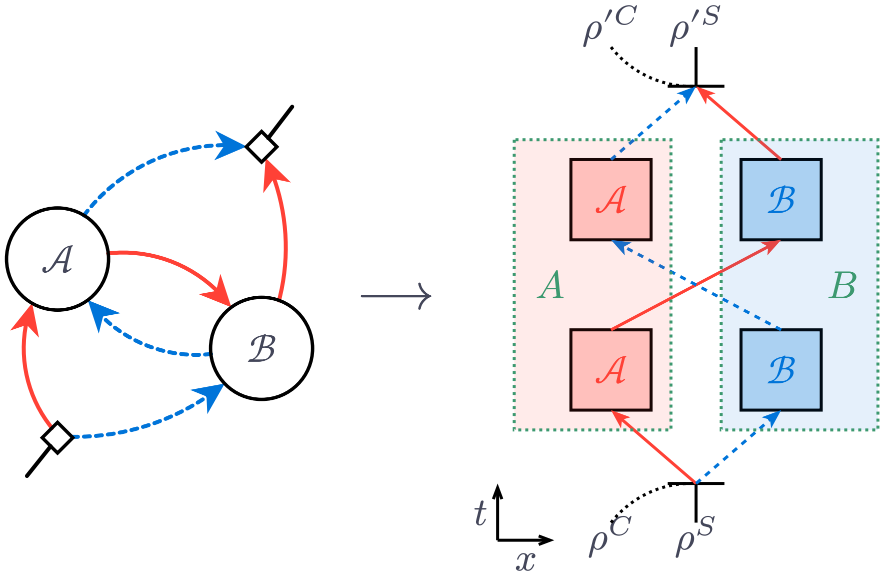
- **Maxwell's demon**

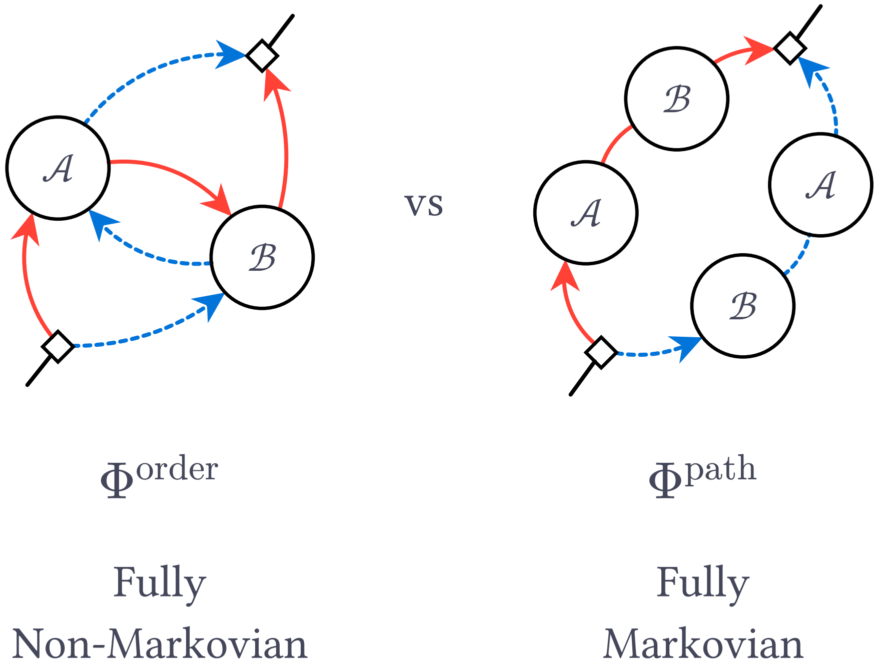
  22. Cheong JW, Pradana A, Chew LY (2026) Enhancing the performance of quantum switch refrigeration with an information pool of qubits. Phys Rev Res 8:13272. https://doi.org/10.1103/3vft-yzlr

Key Takeaways

1. Complex quantum composition of channels can hide **intrinsic *coherent* non-Markovianity**.
2. Which can have **nonnegligible contributions in thermodynamic tasks**.



3. *Coherent* non-Markovianity can offer a **dual explanation to indefinite causal order**.



Related Works

1. 17. Cheong JW, Pradana A, Chew LY (2022) Communication advantage of quantum compositions of channels from non-Markovianity. Phys Rev A 106:52410. https://doi.org/10.1103/PhysRevA.106.052410
2. 26. Karpat G, Çakmak B (2024) Memory in quantum processes with indefinite time direction and causal order. Physical Review A 110:. https://doi.org/10.1103/physreva.110.012446
3. 27. Anand V, Maity AG, Mitra S, Bhattacharya S (2025) Emergent non-Markovianity and dynamical quantification of the quantum switch. Physical Review A 111:. https://doi.org/10.1103/physreva.111.032428

Work	CP-indivisibility	Backflow of information (P-indivisibility)	Time
Cheong et al.	Stinespring dilation	Entanglement monotone, trace distance, quantum relative entropy	Discrete
Karpat et al.	Implied from P-indivisibility	Entanglement monotone, trace distance	Continuous
Anand et al.	RHP measure	Trace distance	Continuous

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Acknowledgements

Thank you for listening!

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Principal Investigator

Dr. Andri PRADANA

Collaborator

Dr. Jian Wei CHEONG

Postdoc (me)

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Nanyang Technological University, Singapore

These slides

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